34 From geosystem to territory: theoretical framework of a GIS-based model of agricultural potential in the Neolithic Pyrenees

Guillem Salvador-Baiges1 (guillem.salvadorbaiges@dainst.de), Ferran Antolin1, 2, Ermengol Gassiot-Ballbè3

Spatial models in archaeology are never theoretically neutral. Behind each algorithm or modelling technique lies a conception of space and the human-environment relationship. In mountain areas, geographically deterministic views have predominated, describing these areas as marginal spaces and their communities in ahistorical terms, always engaged in seasonal livestock farming as a subsidiary activity to the lowlands. This paper focuses on the theoretical framework of an agricultural potential model for the Pyrenees during the Neolithic, developed using FLOS tools (R and QGIS). In other words, it addresses its ontology, premises, hypotheses and limitations. We use the GTL system (Geosystem–Territory–Landscape) as a reference framework, which considers social space as a single reality with three dimensions. The geosystem refers to space as a material reality connected by physicochemical processes; the territory, to its social use; and the landscape, to its symbolic appropriation. We place the study in the territory, understood as the spatial expression of social practices. Thus, social space, observed and analysed from the perspective of the geosystem, defines the conditions of possibility, whereas, analysed through the prism of the territory, it reveals the materialisation of human activities and human labour in space, that is, how space is produced and used at a specific historical moment. From this perspective, we understand modelling as a tool for testing hypotheses. First, we define a specific problem: in this case, whether agriculture was possible in the Pyrenees during the Neolithic for diXerent crops and agricultural strategies. Second, we approximate the past geosystem through climatic, topographic and soil variables and assess its agricultural potential. Finally, we relate this potential to the location of the settlements in order to characterise the territories of the Neolithic Pyrenees. Using FLOS tools makes all these premises explicit and reproducible.